LocateAnything & micro-SAM for EM Segmentation

Three highlights on why they solve different problems — and how they could work together.

July 2026 · Sources: NVIDIA / Hugging Face, arXiv 2605.27365, micro-sam project docs

1 Pixel mask vs. coordinate box — the core mismatch

micro-SAM outputs a **dense pixel mask**; LocateAnything outputs only a **coordinate box** (or point). EM segmentation needs masks, so LocateAnything cannot do it directly. It is also trained purely on natural images, GUIs, OCR and documents — no microscopy — so it cannot even reliably *locate* dense, near-identical EM structures like membranes, mitochondria or vesicles.

> The only useful role: **modify the pipeline.** Fine-tune / train LocateAnything on microscopy data → extract bounding boxes → feed those boxes as prompts to **micro-SAM or SAM 3** to produce the accurate pixel-level segmentation. LocateAnything becomes the box detector; the SAM model remains the mask generator.

2 micro-SAM runs on the SAM 1 backbone

micro-SAM is built on the **original SAM (SAM 1)** — not SAM 2 or SAM 3 — with a custom instance-segmentation decoder added on top. Its encoders are SAM's ViT models (`vit_b` , `vit_l` , `vit_b_em_organelles` , etc.). SAM 2 replaced ViT with **Hiera** encoders and SAM 3 is Meta's Nov-2025 release, so the encoder naming alone confirms SAM 1; SAM 2/3 support is roadmap-only.

3 Sequential/parallel decoding does not apply to micro-SAM

The "sequential vs. parallel box" question only applies to *token-generative* models. LocateAnything is one; micro-SAM is not.

LOCATEANYTHING

Autoregressive VLM (Qwen2.5 decoder) that emits a token sequence. **Parallel Box Decoding** predicts a whole box in one step instead of token-by-token — an optimization of sequential generation.

MICRO-SAM

Not autoregressive — no token sequence at all. A lightweight mask decoder computes the mask in a **single parallel forward pass**. Nothing to "parallelize" because nothing was sequential.

And the arrow is reversed: a box is LocateAnything's **output** but micro-SAM's **input prompt**.

Findings corroborated across cited public sources. Full write-up: locateanything-highlight.html · github.com/thtahamid/3d-segmentation